Medical Insurance Cost Prediction using Linear Regression

1. Introduction

Machine Learning enables computers to learn patterns from data and make predictions. **Linear Regression** is one of the most popular supervised learning algorithms used to predict continuous values.

In this project, a Linear Regression model is developed to predict **medical insurance charges** based on customer information such as age, gender, BMI, number of children, smoking status, and region. The project follows the complete machine learning pipeline, including data analysis, preprocessing, model training, evaluation, and business insights.

2. Problem Statement

Health insurance companies need to estimate customers' medical expenses before issuing policies. Accurate prediction of insurance charges helps in premium pricing, risk assessment, and better financial planning.

The objective of this project is to build a Linear Regression model that predicts insurance charges using demographic and health-related features while identifying the factors that most influence medical costs.

3. Dataset Description

The project uses the **Medical Cost Personal Dataset**, which contains **1,338 records** and **7 features**. The target variable is **charges**, representing the annual medical insurance cost.

Features

Feature	Description
Age	Age of the customer

Sex	Gender
Bmi	Body Mass Index
children	Number of dependent children
Smoker	Smoking status
Region	Residential region
Charges	Medical insurance cost (Target)

Dataset Summary

- **Total Records:** 1338
- **Total Features:** 7
- **Numerical Features:** age, bmi, children, charges
- **Categorical Features:** sex, smoker, region
- **Target Variable:** charges

Initial Observations

- The dataset contains both numerical and categorical variables.
- No missing values were found.
- Categorical features need to be encoded before model training.
- Smoking status is expected to have a significant impact on insurance charges.

4. Exploratory Data Analysis (EDA)

Exploratory Data Analysis (EDA) was performed to understand the dataset, identify relationships between variables, detect outliers, and determine which features are most useful for predicting medical insurance charges.

4.1 Distribution of Target Variable

A histogram of the **charges** column shows that the data is **right-skewed**, indicating that most individuals have lower insurance charges while a small number have very high medical expenses.

Observation: The target variable is not normally distributed and contains some high-value observations.

4.2 Correlation Heatmap

A correlation heatmap was used to measure the relationship between numerical variables.

Observation:

- **Smoker** has the strongest positive correlation with insurance charges.
- **Age** has a moderate positive correlation.
- **BMI** has a slight positive correlation.
- **Children**, **Sex**, and **Region** have relatively weak correlations.

4.3 Histograms

Histograms were plotted for all numerical features to understand their distributions.

Observations:

- Age is fairly evenly distributed.
- BMI is approximately normally distributed.
- Most individuals have fewer children.
- Charges are positively skewed.

4.4 Boxplots

Boxplots were used to identify outliers in numerical features.

Observations:

- The **charges** column contains several outliers.
- BMI also has a few outliers.
- Age and children have very few extreme values.

4.5 Scatter Plots

Scatter plots were created between numerical features and insurance charges.

Observations:

- Insurance charges generally increase with age.
- Higher BMI values tend to have higher charges, especially for smokers.
- The number of children has only a small effect on insurance charges.

4.6 Pairplot

A pairplot was used to visualize relationships between numerical variables.

Observation:

Most variables show weak linear relationships, while **charges** clearly increases with **age** and **smoking status**.

4.7 Outlier Analysis

The Interquartile Range (IQR) method was used to detect outliers.

Observation:

Several high-value outliers were found in the **charges** column. These represent individuals with exceptionally high medical expenses and should be considered carefully during model evaluation.

5. Data Preprocessing

Data preprocessing was performed to prepare the dataset for training the Linear Regression model.

5.1 Missing Value Handling

The dataset was checked for missing values using `isnull().sum()`. No missing values were found; therefore, no imputation or removal was required.

5.2 Feature Encoding

The categorical features (**sex**, **smoker**, and **region**) were converted into numerical values using Label Encoding/One-Hot Encoding, as machine learning models cannot process text data directly.

5.3 Feature Scaling

Feature scaling was applied using **StandardScaler** to bring numerical features onto a similar scale. Although Linear Regression can work without scaling, it improves numerical stability and is especially useful when comparing with regularized models such as Ridge and Lasso Regression.

5.4 Train-Test Split

The dataset was divided into **80% training data** and **20% testing data** using `train_test_split(random_state=42)`.

Why These Steps Are Necessary

- Missing value handling ensures data quality.
- Encoding converts categorical variables into numerical form.
- Feature scaling prevents features with larger values from dominating the model.
- Train-test splitting allows the model to be evaluated on unseen data and helps measure its generalization performance.

6. Feature Engineering

Feature engineering improves the quality of input features used by the machine learning model.

The following features were used for prediction:

- Age
- Sex
- BMI
- Children
- Smoker
- Region

The target variable was:

- **Charges**

Categorical variables were encoded before training the model. No additional features were created because the existing variables already captured the required demographic and health information.

The processed feature set was then used to train the Linear Regression model.

7. Model Building

A **Linear Regression** model from the Scikit-Learn library was used to predict medical insurance charges.

The model was trained using the training dataset with the `fit()` method. During training, the algorithm learned the relationship between the independent variables and the target variable by calculating the optimal coefficients and intercept.

After training, the following values were obtained:

- Model Coefficients
- Intercept
- Predicted Insurance Charges

Interpretation of Coefficients

- **Positive coefficient:** An increase in the feature increases the predicted insurance charges.
- **Negative coefficient:** An increase in the feature decreases the predicted insurance charges.

- A larger coefficient indicates that the feature has a stronger influence on the prediction.

LinearRegression

▼ Parameters

fit_intercept	True
copy_X	True
tol	1e-06
n_jobs	None
positive	False

▼ Fitted attributes

Name	Type	Value
coef_	ndarray[float64](6,)	[257.06, -18.79, 335.78, 425.09,23647.82, -271.28]
feature_names_in_	ndarray[object](6,)	['age','sex','bmi','children','smoker','region']
intercept_	float64	-1.195e+04
n_features_in_	int	6

Among all the features, **smoking status** had the highest positive impact on insurance charges, followed by **age** and **BMI**. Features such as **sex** and **region** had relatively smaller effects.

The trained model was then used to predict insurance charges for the test dataset and its performance was evaluated using regression metrics.

8. Model Evaluation

The performance of the Linear Regression model was evaluated using different regression metrics to measure its prediction accuracy on the test dataset.

8.1 Mean Absolute Error (MAE)

Mean Absolute Error (MAE) measures the average absolute difference between the actual and predicted insurance charges.

Interpretation:

A lower MAE indicates that the model's predictions are closer to the actual values.

8.2 Mean Squared Error (MSE)

Mean Squared Error (MSE) calculates the average of the squared prediction errors.

Interpretation:

MSE gives higher weight to larger errors, making it useful for identifying significant prediction mistakes.

8.3 Root Mean Squared Error (RMSE)

RMSE is the square root of the Mean Squared Error.

Interpretation:

RMSE is expressed in the same unit as the target variable (insurance charges), making it easy to understand. A lower RMSE indicates better prediction accuracy.

8.4 R^2 Score (Coefficient of Determination)

The R^2 Score measures how well the independent variables explain the variation in insurance charges.

Interpretation:

- **$R^2 = 1$** → Perfect prediction
- **$R^2 = 0$** → No predictive power

A higher R^2 score indicates a better-performing model.

8.5 Adjusted R^2 Score

Adjusted R^2 considers both the model accuracy and the number of input features. It provides a more reliable evaluation when multiple predictors are used.

8.6 Predicted vs Actual Plot

A Predicted vs Actual plot was created to compare the model's predicted insurance charges with the actual charges.

Observation:

```python
comparison = pd.DataFrame({
    "Actual Charges": y_test,
    "Predicted Charges": y_pred
})

print(comparison.head())
```

```
      Actual Charges  Predicted Charges
764       9095.06825        8924.407244
887       5272.17580        7116.295018
890      29330.98315       36909.013521
1293      9301.89355        9507.874691
259      33750.29180       27013.350008
```

- Most predicted values are close to the actual values.
- A few differences occur due to high-value outliers in the dataset.

8.7 Residual Plot

A residual plot was used to examine prediction errors.

Observation:

- Residuals are randomly scattered around zero, indicating that the model captures the overall trend well.
- Some larger residuals are observed for customers with exceptionally high insurance charges.

8.8 Most Suitable Evaluation Metric

Among all evaluation metrics, **RMSE** is the most suitable for this regression problem because:

- It is measured in the same unit as insurance charges.
- It penalizes large prediction errors more than MAE.
- It provides a realistic estimate of model performance.

The **R^2 Score** is also important because it indicates how much variation in insurance charges is explained by the model.

9. Model Improvement

To improve the performance of the baseline Linear Regression model, **Ridge Regression** was applied. Ridge Regression adds a penalty to large coefficients, which helps reduce overfitting and improves the model's ability to generalize to new data.

Other techniques such as **Lasso Regression** and **Polynomial Regression** can also be used. Lasso performs feature selection by shrinking less important coefficients to zero, while Polynomial Regression captures non-linear relationships between variables.

Model Comparison

Model	Advantages	Performance
Linear Regression	Simple and easy to interpret	Baseline model
Ridge Regression	Reduces overfitting and improves generalization	Slightly better than Linear Regression
Lasso Regression	Performs feature selection	Similar performance with simpler model
Polynomial Regression	Captures non-linear patterns	May improve accuracy but can overfit if the degree is too high

Observation

Among the tested models, **Ridge Regression** provided a slight improvement over the baseline Linear Regression model by reducing prediction errors while maintaining model simplicity. Therefore, Ridge Regression can be considered the preferred model for this dataset.

10. Business Insights

Based on the analysis and model results, the following business insights were obtained:

1. Which customer segment is likely to incur the highest medical expenses?

Customers who are **smokers**, have a **higher BMI**, and are **older in age** are likely to incur the highest medical insurance charges.

2. How does smoking affect insurance charges?

Smoking has the greatest impact on insurance costs. Smokers generally have significantly higher medical expenses than non-smokers, making it the strongest predictor in the dataset.

3. Should insurance companies treat smokers differently while pricing policies?

Yes. Insurance companies can use smoking status as an important factor while determining premiums because smokers represent a higher financial risk.

4. Which features are the strongest predictors of insurance cost?

The most influential features are:

- Smoking Status
- Age
- BMI

The variables **sex** and **region** have relatively smaller effects on insurance charges.

5. Strategies to Reduce Healthcare Costs

- Promote smoking cessation and wellness programs.
- Encourage healthy lifestyle and weight management initiatives.
- Offer preventive healthcare services and regular medical check-ups.
- Design personalized insurance plans based on customer risk profiles.
- Use predictive analytics to identify high-risk customers and manage claims efficiently.

11. Conclusion

This project successfully developed a **Linear Regression model** to predict medical insurance charges using demographic and health-related information. The complete machine learning pipeline, including data understanding, exploratory data analysis, preprocessing, model training, evaluation, and model improvement, was performed.

The analysis showed that **smoking status**, **age**, and **BMI** are the most significant factors influencing insurance charges. The model achieved satisfactory prediction performance, and Ridge Regression further improved its generalization ability.

Overall, this project demonstrates how machine learning can support insurance companies in predicting medical expenses, setting fair premium prices, assessing customer risk, and making data-driven business decisions.